Econ 5551

Design-Based Inference, Random. Tests

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Two Frameworks for Inference in RCTs

1. **Design-based (randomization-based) inference:**
   - *Only* source of randomness is the treatment assignment mechanism, *known* by design.
   - Inference is conditional on the realized sample; no reference to a population outside the sample.
2. **Sampling-based (superpopulation) inference:**
   - Units are modeled as drawn i.i.d. from a hypothetical superpopulation, introducing *additional* randomness beyond the assignment mechanism.
   - Inference targets population parameters, e.g. $E\left[Y_1 - Y_0\right]$ over the superpopulation.

Two key distinctions:

1. **Source of randomness:** assignment mechanism *only* (design-based) vs. assignment mechanism *plus* sampling from a population (superpopulation).
2. **Inferential targets:** sample quantities, e.g. sample ATE $\frac{1}{N}\sum_i\left(Y_{1,i} - Y_{0,i}\right)$, or sharp null $Y_{1,i} = Y_{0,i}$ for each i (design-based); vs. population quantities, e.g. population ATE $E\left[Y_1 - Y_0\right]$, or null $E\left[Y_1 - Y_0\right] = 0$ (superpopulation).

Design-Based Inference: Setup

Consider an experiment with N units, N_1 assigned to treatment.

- Each unit i has *fixed* potential outcomes $\left(Y_{0,i}, Y_{1,i}\right)$.
- Treatment is assigned by a *known* randomization procedure, producing the random assignment vector $\mathbf{D} = \left(D_1, \ldots, D_N\right)$ with known distribution $\Pr[\mathbf{D} = \mathbf{d}]$.
  - E.g., for a completely randomized experiment: $\Pr[\mathbf{D} = \mathbf{d}] = 1/\binom{N}{N_1}$ for all $\mathbf{d}$ with $\sum_i d_i = N_1$.
- Observed outcome: $Y_i = Y_{0,i} + D_i\left(Y_{1,i} - Y_{0,i}\right)$.
- **Key idea: D** is the *only* stochastic element. We can use this *known* distribution to construct reference distributions for test statistics and to derive sampling properties of estimators—without modeling outcomes as random draws from a population.
- Seminal references: Fisher (1935), Neyman (1923/1990).
- Textbook reference: Chapters 5–6 of Imbens and Rubin (2015).

Contrasting the Two Frameworks

	Design-Based	**Superpopulation**
Source of randomness	Assignment mechanism *only*	Assignment mechanism *plus* sampling from population
Potential outcomes	Fixed	Random
Target	Finite-sample quantities, e.g. $\frac{1}{N}\sum_i \left(Y_{1,i} - Y_{0,i}\right)$	Population quantities, e.g. $E\left[Y_1 - Y_0\right]$
Assumptions	Known assignment mechanism	Distributional / sampling assumptions (may also exploit assignment mechanism)

Contrasting the Two Frameworks

- The two frameworks often give similar answers in large samples, but differ conceptually and can differ in finite samples:
  - Superpopulation inference is typically conservative from the design-based perspective, because it accounts for sampling variation that the design-based approach conditions away.
  - Conversely, from the superpopulation perspective, design-based inference can be misleadingly precise, because it ignores the randomness of which units end up in the study and targets only the sample ATE rather than the population ATE.
- See, e.g., Section 3 of Bai, Shaikh, and Tabord-Meehan (2025) for comparison of approaches.
- In either framework, the known randomization procedure plays a key role, with results depending on particular randomization procedure.

Example Randomization Procedure: CRE

In a Complete Randomized Experiment (CRE), a fixed number N_1 of N units are randomly selected for treatment, with all $\binom{N}{N_1}$ assignments equally likely, so that:

$$\Pr[\mathbf{D} = \mathbf{d}] = 1 \Big/ \binom{N}{N_1} \text{ for all } \mathbf{d} \text{ with } \sum_i d_i = N_1.$$

Complete randomization implies D_i, D_j are negatively correlated:

$$\operatorname{Corr}(D_i, D_j) = -\frac{1}{N-1} \quad (i \neq j).$$

Intuition: Since fixed number of units selected for treatment, unit i selected makes it less likely that unit j will be selected.

Today

For Complete Randomized Experiment:

1. Consider expected value and variance of mean difference estimator from both design-based and superpopulation perspectives;
2. Fisher-style randomization test of the sharp null from design based perspective.

We will finish by considering re-randomization and post-randomization adjustments, and covariate-adaptive experimental designs such as block stratification.

Mean Difference Estimator: Setup

Consider a completely randomized experiment with N units, N_1 assigned to treatment, $N_0 = N - N_1$ to control. Here N_1 and N_0 are fixed ex ante.

- The *mean difference estimator* (difference in means) is:

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0,$$

where

$$\bar{Y}_1 = \frac{1}{N_1}\sum_{i=1}^{N} D_i Y_i = \frac{1}{N_1}\sum_{i=1}^{N} D_i Y_{1,i},$$

$$\bar{Y}_0 = \frac{1}{N_0}\sum_{i=1}^{N} (1 - D_i) Y_i = \frac{1}{N_0}\sum_{i=1}^{N} (1 - D_i) Y_{0,i}.$$

Mean Difference Estimator: Setup (cont'd)

- Define finite-population average treatment effect as:

$$\tau_{\text{fs}} = \frac{1}{N}\sum_{i}\left(Y_{1,i} - Y_{0,i}\right) = \bar{Y}(1) - \bar{Y}(0),$$

where

$$\bar{Y}(1) = \frac{1}{N}\sum_{i=1}^{N} Y_{1,i}, \quad \bar{Y}(0) = \frac{1}{N}\sum_{i=1}^{N} Y_{0,i}.$$

- Define finite-population variances as:

$$S_d^2 = \frac{1}{N-1}\sum_{i=1}^{N}\left(Y_{d,i} - \bar{Y}(d)\right)^2 \text{ for } d \in \{0, 1\},$$

$$S_\tau^2 = \frac{1}{N-1}\sum_{i=1}^{N}\left[\left(Y_{1,i} - Y_{0,i}\right) - \tau_{\text{fs}}\right]^2.$$

Design-Based: Unbiasedness

Proposition (Neyman, 1923)

Under complete randomization, $E_{\mathbf{D}}\left[\hat{\tau}\right] = \tau_{\text{fs}}$.

Proof. Under complete randomization, each unit is equally likely to be assigned to treatment:

$$E\left[D_i\right] = \frac{N_1}{N} \quad \text{for all } i.$$

Since potential outcomes are fixed:

$$E\left[\overline{Y}_1\right] = E\left[\frac{1}{N_1}\sum_i D_i Y_{1,i}\right] = \frac{1}{N_1}\sum_i E\left[D_i\right] Y_{1,i} = \frac{1}{N_1}\sum_i \frac{N_1}{N} Y_{1,i} = \frac{1}{N}\sum_i Y_{1,i} = \overline{Y}(1).$$

By the same argument, $E\left[\overline{Y}_0\right] = \overline{Y}(0)$. Therefore,

$$E\left[\hat{\tau}\right] = \overline{Y}(1) - \overline{Y}(0) = \frac{1}{N}\sum_i \left(Y_{1,i} - Y_{0,i}\right) = \tau_{\text{fs}}. \qquad \square$$

Design-Based: Variance (Neyman)

Proposition (Neyman, 1923)

Under complete randomization:

$$\mathrm{Var}_{\mathbf{D}}\left(\hat{\tau}\right) = \frac{S_1^2}{N_1} + \frac{S_0^2}{N_0} - \frac{S_\tau^2}{N}.$$

Recall: $S_\tau^2 = \frac{1}{N-1}\sum_{i=1}^{N}\left[\left(Y_{1,i} - Y_{0,i}\right) - \tau_{\text{fs}}\right]^2$.

- The term $-S_\tau^2/N$ involves the variance of the *unit-level* treatment effects $\tau_i = Y_{1,i} - Y_{0,i}$—which is unobservable (we never see both potential outcomes for the same unit).
- Therefore $\mathrm{Var}_{\mathbf{D}}\left(\hat{\tau}\right)$ is not directly estimable without further assumptions.

Design-Based: Variance (Proof)

Proof sketch. The key ingredients under complete randomization (sampling N_1 of N units without replacement):

$$\mathrm{Var}\big(D_i\big) = \frac{N_1}{N} \cdot \frac{N_0}{N}, \qquad \mathrm{Cov}\big(D_i, D_j\big) = -\frac{N_1 N_0}{N^2\big(N-1\big)} \quad \big(i \neq j\big).$$

- Write $\hat{\tau} = \frac{1}{N_1}\sum_i D_i Y_{1,i} - \frac{1}{N_0}\sum_i \big(1 - D_i\big) Y_{0,i}$ as a linear function of the D_i's applied to fixed potential outcomes.
- Compute $\mathrm{Var}\big(\hat{\tau}\big)$ using $\mathrm{Var}\big(D_i\big)$ and $\mathrm{Cov}\big(D_i, D_j\big)$, collecting terms involving S_1^2, S_0^2, and S_τ^2.

Design-Based: Variance (cont'd)

Proof (completing the variance calculation). Write

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0 = \sum_{i=1}^{N} c_i D_i + r,$$

where c_i and r are functions of the fixed potential outcomes. We have

$$\begin{aligned}\hat{\tau} &= \frac{1}{N_1}\sum_i D_i Y_{1,i} - \frac{1}{N_0}\sum_i (1 - D_i) Y_{0,i} \\ &= \sum_i D_i \left(\frac{Y_{1,i}}{N_1} + \frac{Y_{0,i}}{N_0} \right) - \frac{1}{N_0}\sum_i Y_{0,i},\end{aligned}$$

so $c_i = Y_{1,i}/N_1 + Y_{0,i}/N_0$, and $r = -\frac{1}{N_0}\sum_i Y_{0,i}$.

Design-Based: Variance (cont'd)

Proof (completing the variance calculation). Applying

$$\text{Var}\Big(\sum_i c_i D_i\Big) = \sum_i c_i^2 \text{Var}\big(D_i\big) + \sum_{i \neq j} c_i c_j \text{Cov}\big(D_i, D_j\big).$$

and substituting the expressions for $\text{Var}\big(D_i\big)$, $\text{Cov}\big(D_i, D_j\big)$, one can show:

$$\text{Var}_{\mathbf{D}}\big(\hat{\tau}\big) = \frac{S_1^2}{N_1} + \frac{S_0^2}{N_0} - \frac{S_\tau^2}{N}. \qquad \square$$

Design-Based: Neyman's Conservative Variance Estimator

- Since S_τ^2 is not identifiable (requires observing both $Y_{0,i}$ and $Y_{1,i}$ for the same unit), Neyman proposed the *conservative* variance estimator:

$$\hat{V}_{\text{Neyman}} = \frac{s_1^2}{N_1} + \frac{s_0^2}{N_0},$$

  where $s_d^2 = \frac{1}{N_d - 1} \sum_{i:D_i=d} \left(Y_i - \bar{Y}_d\right)^2$ is the sample variance in group d.
- Under complete randomization, $E\left[s_d^2\right] = S_d^2$, so:

$$E\left[\hat{V}_{\text{Neyman}}\right] = \frac{S_1^2}{N_1} + \frac{S_0^2}{N_0} = \text{Var}_{\mathbf{D}}\left(\hat{\tau}\right) + \frac{S_\tau^2}{N} \geq \text{Var}_{\mathbf{D}}\left(\hat{\tau}\right).$$

Design-Based: Neyman's Conservative Variance Estimator

$$E\left[\hat{V}_{\text{Neyman}}\right] = \frac{S_1^2}{N_1} + \frac{S_0^2}{N_0} = \text{Var}_{\mathbf{D}}\left(\hat{\tau}\right) + \frac{S_\tau^2}{N} \geq \text{Var}_{\mathbf{D}}\left(\hat{\tau}\right).$$

- The Neyman variance estimator results in *conservative* inference: it overestimates the true variance by $S_\tau^2 / N \geq 0$, with equality iff treatment effects are constant ($\tau_i = \tau_{\text{fs}}$ for all i).
- Aronow, Green, and Lee (2014) show that one can consistently estimate *sharp bounds* on $\text{Var}_{\mathbf{D}}\left(\hat{\tau}\right)$, yielding tighter (asymptotically narrowest) conservative confidence intervals than those based on $\hat{V}_{\text{Neyman}}$.

Superpopulation: Unbiasedness

Now model $\left(Y_{0,i}, Y_{1,i}\right)$ as i.i.d. draws from a superpopulation with $E\left[Y_d\right] = \mu_d$ and $\text{Var}\left(Y_d\right) = \sigma_d^2$ for $d \in \left\{0, 1\right\}$.

- Even though $\left\{\left(Y_{0,i}, Y_{1,i}\right)\right\}_{i=1}^N$ are $i.i.d.$, $\left\{\left(D_i, Y_i\right)\right\}_{i=1}^N$ are not i.i.d. because of experimental design.

Define the population ATE as:

$$\tau = \mu_1 - \mu_0 = E\left[Y_1 - Y_0\right].$$

Proposition

Under random sampling from the superpopulation and complete randomization, $E\left[\hat{\tau}\right] = \tau$.

Superpopulation: Unbiasedness (Proof)

Proof. Taking expectation over both sampling and assignment:

$$E\left[\overline{Y}_1\right] = E\left[\frac{1}{N_1}\sum_i D_i Y_{1,i}\right] = \frac{1}{N_1}\sum_i E\left[D_i Y_{1,i}\right] = \frac{1}{N_1}\sum_i E\left[D_i\right] E\left[Y_{1,i}\right].$$

The last equality uses independence of D_i and $Y_{1,i}$ (treatment assignment is independent of potential outcomes, by design). Since $E\left[D_i\right] = N_1/N$ and $E\left[Y_{1,i}\right] = \mu_1$:

$$E\left[\overline{Y}_1\right] = \frac{1}{N_1} \cdot N \cdot \frac{N_1}{N} \cdot \mu_1 = \mu_1.$$

By the same argument, $E\left[\overline{Y}_0\right] = \mu_0$, so $E\left[\hat{\tau}\right] = \mu_1 - \mu_0 = \tau$. $\square$

Superpopulation: Variance

Proposition

Under random sampling from the superpopulation and complete randomization:

$$\mathrm{Var}_{\mathrm{sp}}(\hat{\tau}) = \frac{\sigma_1^2}{N_1} + \frac{\sigma_0^2}{N_0}.$$

- Compared to the design-based variance $\frac{S_1^2}{N_1} + \frac{S_0^2}{N_0} - \frac{S_\tau^2}{N}$, the superpopulation variance has no S_τ^2 / N correction—the covariance term vanishes because the superpopulation approach accounts for sampling variation across different realizations of the N units.
- Note: the superpopulation variance σ_d^2 / N_d uses population variances σ_d^2, while the design-based variance uses finite-sample variances S_d^2.

Superpopulation: Variance (Proof)

Proof. By the law of total variance:

$$\mathrm{Var}_{\mathrm{sp}}\big(\bar{Y}_1\big) = E_S\Big[\mathrm{Var}_{\mathbf{D}}\big(\bar{Y}_1 \mid S\big)\Big] + \mathrm{Var}_S\Big(E_{\mathbf{D}}\big[\bar{Y}_1 \mid S\big]\Big).$$

Given the sample S, $\mathrm{Var}_{\mathbf{D}}\big(\bar{Y}_1 \mid S\big) = \frac{N_0}{N_1 N(N-1)} \sum_i \big(Y_{1,i} - \bar{Y}(1)\big)^2$ (hypergeometric variance). Taking expectations over S and using $E_S\left[\frac{1}{N-1}\sum_i \big(Y_{1,i} - \bar{Y}(1)\big)^2\right] = \sigma_1^2$: the first term contributes $\frac{N_0}{N_1 N}\sigma_1^2$.
The second term: $E_{\mathbf{D}}\big[\bar{Y}_1 \mid S\big] = \bar{Y}(1)$, and $\mathrm{Var}_S\big(\bar{Y}(1)\big) = \sigma_1^2/N$. So:

$$\mathrm{Var}_{\mathrm{sp}}\big(\bar{Y}_1\big) = \frac{N_0}{N_1 N}\sigma_1^2 + \frac{\sigma_1^2}{N} = \frac{\sigma_1^2}{N_1}.$$

Analogous calculation for $\bar{Y}_0$, plus independence of $\bar{Y}_1$ and $\bar{Y}_0$ conditional on treatment sizes, gives the result. $\square$

Comparing Variances Across Frameworks

- **Design-based:** $\text{Var}_{\mathbf{D}}(\hat{\tau}) = \dfrac{S_1^2}{N_1} + \dfrac{S_0^2}{N_0} - \dfrac{S_\tau^2}{N}$.
- **Superpopulation:** $\text{Var}_{\text{sp}}(\hat{\tau}) = \dfrac{\sigma_1^2}{N_1} + \dfrac{\sigma_0^2}{N_0}$.
- Taking expectation of the design-based variance over the superpopulation:

$$E_S\left[\frac{S_d^2}{N_d}\right] = \frac{\sigma_d^2}{N_d}, \qquad E_S\left[\frac{S_\tau^2}{N}\right] = \frac{\sigma_\tau^2}{N},$$

where $\sigma_\tau^2 = \text{Var}\big(Y_{1,i} - Y_{0,i}\big)$. So:

$$E_S\big[\text{Var}_{\mathbf{D}}(\hat{\tau})\big] = \frac{\sigma_1^2}{N_1} + \frac{\sigma_0^2}{N_0} - \frac{\sigma_\tau^2}{N} = \text{Var}_{\text{sp}}(\hat{\tau}) - \frac{\sigma_\tau^2}{N}.$$

- Since $\sigma_\tau^2 \geq 0$, the superpopulation variance is weakly larger:

$$\text{Var}_{\text{sp}}(\hat{\tau}) = E_S\big[\text{Var}_{\mathbf{D}}(\hat{\tau})\big] + \frac{\sigma_\tau^2}{N} \;\geq\; E_S\big[\text{Var}_{\mathbf{D}}(\hat{\tau})\big],$$

Comparing Variances: Intuition

- The gap σ_τ^2/N equals the variance of the sample ATE across different draws of N units from the superpopulation:

$$\mathrm{Var}_S(\tau_{\mathrm{fs}}) = \mathrm{Var}_S\left(\frac{1}{N}\sum_i (Y_{1,i} - Y_{0,i})\right) = \frac{\sigma_\tau^2}{N}.$$

- **Design-based** inference conditions on the realized sample, so only captures variability from the randomization mechanism—not from which N units ended up in the sample.
- **Superpopulation** inference targets $\tau = E[Y_1 - Y_0]$, and accounts for *both* randomization variation and sampling variation.
- Hence $\hat{V}_{\mathrm{Neyman}} = s_1^2/N_1 + s_0^2/N_0$ is unbiased for the *superpopulation* variance $\sigma_1^2/N_1 + \sigma_0^2/N_0$, which is why it is *conservative* for the design-based variance.

Variance Decomposition Across Frameworks

To summarize, the results on the preceding slides imply the following decomposition. From the **superpopulation** perspective, the total variance of $\bar{Y}_1 - \bar{Y}_0$ in a completely randomized experiment decomposes as:

$$\mathrm{Var}_{\mathrm{sp}}\left(\bar{Y}_1 - \bar{Y}_0\right) = \underbrace{E_S\left[\mathrm{Var}_{\mathbf{D}}\left(\bar{Y}_1 - \bar{Y}_0 \mid S\right)\right]}_{\substack{\text{expected design-based variance,} \\ \text{averaged over random samples}}} + \underbrace{\mathrm{Var}_S\left(\frac{1}{N}\sum_i\left(Y_{1,i} - Y_{0,i}\right)\right)}_{\substack{\text{variation from which} \\ \text{units are in the sample}}}$$

- **Design-based inference** conditions on the realized sample, so it captures only the first term. From this perspective, the superpopulation variance is "too big"—it includes sampling variation that, conditional on our sample, is not present.
- **Superpopulation inference** targets the population ATE, $E\left[Y_1 - Y_0\right]$. From this perspective, design-based CIs are "too short"—they miss the second term, which reflects the fact that a different draw of N units from the population would yield a different sample ATE, $\frac{1}{N}\sum_i\left(Y_{1,i} - Y_{0,i}\right)$.

Example: PROGRESA

- $Y_{0,i}$ denote an indicator variable for whether child i would be enrolled in school without PROGRESA.
- $Y_{1,i}$ denote an indicator variable for whether child i would be enrolled in school with PROGRESA.
- D_i denote a dummy variable for whether child i was treated.
- $Y_i = Y_{0,i} + D_i\left(Y_{1,i} - Y_{0,i}\right)$ is observed school enrollment for child i.
- We observe $Y_{0,i}$ or $Y_{1,i}$ for each child, but never both.

Example: PROGRESA

- $Y_{1,i} - Y_{0,i}$ is effect of PROGRESA on child i's enrollment.
- $\frac{1}{N}\sum_{i=1}^{N}\left(Y_{1,i} - Y_{0,i}\right)$ is the (sample) average treatment effect of PROGRESA on school enrollment.

What if wished to test that PROGRESA had no effect?

1. Could test $H_0 : Y_{1,i} = Y_{0,i}$ for all i, vs $H_1 : Y_{1,i} \neq Y_{0,i}$ for some i.
   (null of no effect on any child, vs alternative on an effect on at least one child)
2. Could test $H_0 : \frac{1}{N}\sum_i\left(Y_{1,i} - Y_{0,i}\right) = 0$, vs $H_1 : \frac{1}{N}\sum_i\left(Y_{1,i} - Y_{0,i}\right) \neq 0$.
   (null of no effect on average, vs alternative on nonzero average effect)

Example: PROGRESA

- $Y_{1,i} - Y_{0,i}$ is effect of PROGRESA on child i's enrollment.
- $\frac{1}{N}\sum_{i=1}^{N}\left(Y_{1,i} - Y_{0,i}\right)$ is the (sample) average treatment effect of PROGRESA on school enrollment.

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2. Could test $H_0 : \frac{1}{N}\sum_i\left(Y_{1,i} - Y_{0,i}\right) = 0$, vs $H_1 : \frac{1}{N}\sum_i\left(Y_{1,i} - Y_{0,i}\right) \neq 0$.
   (null of no effect on average, vs alternative on nonzero average effect)

We will now consider randomization inference to perform a finite-sample test of 1. Under complete randomization, this coincides with a permutation test, but the validity comes from the known assignment mechanism rather than an exchangeability assumption—a distinction that matters under more complex designs.

Sharp Null Hypothesis of No Effect

Definition (Sharp Null of No Effect)

$H_0 : Y_{1,i} = Y_{0,i}$ for all i.

- Called the "*sharp null hypothesis*" (sometimes "*exact null hypothesis*") because we know the values of both $Y_{0,i}$ and $Y_{1,i}$ *under the null*.
- The null that $\frac{1}{N}\sum_i \left(Y_{1,i} - Y_{0,i}\right) = 0$ is a weaker hypothesis.
  - No effect on any i implies no effect on average.
  - No effect on average does not imply no effect on any i.
- Exploiting random assignment in a randomized controlled trial (RCT), we can test the sharp null using a permutation test, called a randomization test in this context.

Example: PROGRESA With Village as Unit

For ease of exposition, consider PROGRESA taking unit of observation to be the village, aggregating data to village level.

- Village level corresponds to level of randomization;
- Will later consider individuals as unit while still recognizing that randomization was at village level (*cluster RCT*).

Example: PROGRESA With Village as Unit

For ease of exposition, consider PROGRESA taking unit of observation to be the village, aggregating data to village level.

- Village level corresponds to level of randomization;
- Will later consider individuals as unit while still recognizing that randomization was at village level (*cluster RCT*).

Aggregating to village level:

- $Y_{0,i}$ denote average enrollment for children in village i without PROGRESA;
- $Y_{1,i}$ denote average enrollment for children in village i with PROGRESA;
- D_i denote dummy variable for village i being assigned to treatment;
- $Y_i = Y_{0,i} + D_i\big(Y_{1,i} - Y_{0,i}\big)$ is observed average enrollment for children in village i.

PROGRESA, 5 Villages

- For ease of exposition, lets consider just 5 villages, will return to later consider all 491 villages.
- Consider villages 1, 2, 3, 4, and 13 from your PROGRESA data set.
- Relabel village number 13 to be number 5 for ease of exposition.

```
# Recall sooloca is village id, dfPost is our dataframe
> dfpost5  <- subset(dfPost, sooloca %in% c(1, 2, 3, 4, 13))
> dfpost5$sooloca <-
     ifelse(dfpost5$sooloca==13,5,dfpost5$sooloca)
```

PROGRESA, 5 Villages

	Potential Outcomes		Observed Variables	
i	$Y_{0,i}$	$Y_{1,i}$	Y_i	D_i
1				
2				
3				
4				
5				

Filling in table for these 5 villages while aggregating data to village level. . . .

PROGRESA, 5 Villages

	Potential Outcomes		Observed Variables	
i	$Y_{0,i}$	$Y_{1,i}$	Y_i	D_i
1				1
2				1
3				0
4				1
5				0

Treatment variable D_i is *treat*, assigned at village level:

```
> with(dfpost5,tapply(treat, sooloca, mean))
1 2 3 4 5
1 1 0 1 0
```

PROGRESA, 5 Villages

	Potential Outcomes		Observed Variables	
i	$Y_{0,i}$	$Y_{1,i}$	Y_i	D_i
1			0.65	1
2			0.62	1
3			0.81	0
4			0.88	1
5			0.60	0

Outcome Y_i is *school*, school enrollment:

```
> with(dfpost5,round(tapply(school, sooloca, mean),2))
   1    2    3    4    5
0.65 0.62 0.81 0.88 0.60
```

PROGRESA, 5 Villages

	Potential Outcomes		Observed Variables	
i	$Y_{0,i}$	$Y_{1,i}$	Y_i	D_i
1	?	0.65	0.65	1
2	?	0.62	0.62	1
3			0.81	0
4	?	0.88	0.88	1
5			0.60	0

- If $D_i = 1$,
  - $Y_{1,i} = Y_i$;
  - $Y_{0,i} =$ Unknown.

PROGRESA, 5 Villages

	Potential Outcomes		Observed Variables	
i	$Y_{0,i}$	$Y_{1,i}$	Y_i	D_i
1	?	0.65	0.65	1
2	?	0.62	0.62	1
3	0.81	?	0.81	0
4	?	0.88	0.88	1
5	0.60	?	0.60	0

- If $D_i = 1$,
  - $Y_{1,i} = Y_i$;
  - $Y_{0,i} =$ Unknown.
- If $D_i = 0$,
  - $Y_{1,i} =$ Unknown;
  - $Y_{0,i} = Y_i$.

PROGRESA, 5 Villages

	Potential Outcomes		Observed Variables	
i	$Y_{0,i}$	$Y_{1,i}$	Y_i	D_i
1	?	0.65	0.65	1
2	?	0.62	0.62	1
3	0.81	?	0.81	0
4	?	0.88	0.88	1
5	0.60	?	0.60	0

- If $D_i = 1$,
  - $Y_{1,i} = Y_i$;
  - $Y_{0,i} =$ Unknown.
- If $D_i = 0$,
  - $Y_{1,i} =$ Unknown;
  - $Y_{0,i} = Y_i$.

For each unit:

- observe either $Y_{0,i}$ or $Y_{1,i}$, never both,
- do not know $Y_{1,i} - Y_{0,i}$ for any unit.

PROGRESA, 5 Villages

	Potential Outcomes		Observed Variables	
i	$Y_{0,i}$	$Y_{1,i}$	Y_i	D_i
1	?	0.65	0.65	1
2	?	0.62	0.62	1
3	0.81	?	0.81	0
4	?	0.88	0.88	1
5	0.60	?	0.60	0

- Estimate *Average Treatment Effect* (ATE) by $\bar{Y}_1 - \bar{Y}_0$, where $\bar{Y}_1$ is sample mean of Y_i among treated observations, $\bar{Y}_0$ sample mean of Y_i among control observations:

$$\bar{Y}_1 - \bar{Y}_0 = \frac{0.65 + 0.62 + 0.88}{3} - \frac{0.81 + 0.60}{2} = 0.0117.$$

PROGRESA, 5 Villages

	Potential Outcomes		Observed Variables	
i	$Y_{0,i}$	$Y_{1,i}$	Y_i	D_i
1	?	0.65	0.65	1
2	?	0.62	0.62	1
3	0.81	?	0.81	0
4	?	0.88	0.88	1
5	0.60	?	0.60	0

- Consider *sharp null* of no treatment effect on any unit:
  $H_0 : Y_{1i} = Y_{0i}$ for all i, vs $H_1 : Y_{1i} \neq Y_{0i}$ for some i.
- Sharp null implies that $\frac{1}{N}\sum_i (Y_{1,i} - Y_{0,i}) = 0$.
- Possible for $\frac{1}{N}\sum_i (Y_{1,i} - Y_{0,i}) = 0$ but for sharp null to be false.

PROGRESA, 5 Villages

Under the Sharp Null

	Potential Outcomes		Observed Variables	
i	$Y_{0,i}$	$Y_{1,i}$	Y_i	D_i
1	(0.65)	0.65	0.65	1
2	(0.62)	0.62	0.62	1
3	0.81	(0.81)	0.81	0
4	(0.88)	0.88	0.88	1
5	0.60	(0.60)	0.60	0

- Consider *sharp null* of no treatment effect on any unit: $H_0 : Y_{1i} = Y_{0i}$ for all i.
- Under sharp null hypothesis, we can fill in missing elements of table.
- We can exploit randomization procedure and that we know both $Y_{0,i}$ and $Y_{1,i}$ under the null to test the null (randomization test).

Randomization Inference

To perform test, we need to:

1. State null hypothesis and alternative,
   - $H_0 : Y_{1i} = Y_{0i}$ for all i, vs $H_1 : Y_{1i} \neq Y_{0i}$ for some i.
2. Choose a test statistic, larger values providing more evidence against the null,
   - Take test statistic to be $T = \left|\overline{Y}_1 - \overline{Y}_0\right|$. (*Why appropriate*?)
   - Can consider other test statistics . . .
3. Calculate or approximate distribution of test statistic under the null hypothesis to construct p-values, then reject or not reject at desired significance level.

Randomization Inference: Randomization Distribution of T

How to calculate distribution of test statistic?

- Condition on our combined sample of treated and control observations.
- Under sharp null hypothesis, for each village, we would have observed the same outcome whether assigned to treated sample or to control sample.
- Exploit randomness from *known randomization procedure* for RCT to obtain (conditional) randomization distribution of test statistic.

Randomization Inference: Randomization Distribution of T (cont'd)

In our example, if randomization procedure was to randomly draw 3 out of 5 villages for treatment:

- We have $\binom{5}{3} = 10$ possible ways to assign treatment to 3 out of 5 villages, each of the 10 ways equally likely. (Can generalize to more complicated randomization procedures).

Randomization Inference: Randomization Distribution of T (cont'd)

In our example, if randomization procedure was to randomly draw 3 out of 5 villages for treatment:

- We have $\binom{5}{3} = 10$ possible ways to assign treatment to 3 out of 5 villages, each of the 10 ways equally likely. (Can generalize to more complicated randomization procedures).

- Let $T_1, \ldots T_{10}$ denote the corresponding values of the test statistic under the null, each value equally likely given the randomization procedure, so that, for any t,

$$\Pr[T \geq t] = \frac{1}{10} \sum_{j=1}^{10} \mathbf{1}[T_j \geq t],$$

proportion of test statistics at least as large as t.

Randomization Inference: p-value

- In our actual sample we observe $T = 0.012$.
  - p-value: Under null, how likely are we to observe a test statistic this large or larger?
  - We can calculate conditional, exact p-value as $\frac{1}{10}\sum_{j=1}^{10} \mathbf{1}\big[T_j \geq 0.012\big]$, proportion of test statistics at least as large as observed test statistic.
  - For an α level test, reject sharp null if p-value less than or equal to α.
- Permutation test, called randomization test in this context.

Randomization Test: General Framework

Consider a completely randomized experiment: N_1 of N units assigned to treatment.

Definition (Randomization Distribution)

Under the sharp null $H_0 : Y_{1,i} = Y_{0,i}$ for all i, the observed outcome Y_i is invariant to D_i, so the randomization distribution of any test statistic $T(\mathbf{Y}, \mathbf{D})$ is determined entirely by the known distribution of $\mathbf{D}$:

$$\Pr_{H_0}[T \geq t] = \frac{1}{\binom{N}{N_1}} \sum_{\mathbf{d} \in \Omega} \mathbf{1}[T(\mathbf{Y}, \mathbf{d}) \geq t],$$

where $\Omega = \{\mathbf{d} \in \{0,1\}^N : \sum_i d_i = N_1\}$.

Randomization Test: General Framework

Consider a completely randomized experiment: N_1 of N units assigned to treatment.

Definition (Randomization Distribution)

Under the sharp null $H_0 : Y_{1,i} = Y_{0,i}$ for all i, the observed outcome Y_i is invariant to D_i, so the randomization distribution of any test statistic $T(\mathbf{Y}, \mathbf{D})$ is determined entirely by the known distribution of $\mathbf{D}$:

$$\Pr_{H_0}[T \geq t] = \frac{1}{\binom{N}{N_1}} \sum_{\mathbf{d} \in \Omega} \mathbf{1}[T(\mathbf{Y}, \mathbf{d}) \geq t],$$

where $\Omega = \{\mathbf{d} \in \{0,1\}^N : \sum_i d_i = N_1\}$.

- Under H_0: each unit's outcome is Y_i regardless of assignment, so the vector $\mathbf{Y}$ is fixed.
- The distribution of T is induced solely by the equiprobable permutations of $\mathbf{D}$.
- No distributional assumptions on $\mathbf{Y}$; no asymptotic approximation.

Randomization Test: p-value

Definition (Randomization p-value)

$$p = \Pr_{H_0}[T \geq T^{\text{obs}}] = \frac{1}{\binom{N}{N_1}} \sum_{\mathbf{d} \in \Omega} \mathbf{1}[T(\mathbf{Y}, \mathbf{d}) \geq T^{\text{obs}}],$$

where $\Omega = \{\mathbf{d} \in \{0,1\}^N : \sum_i d_i = N_1\}$.

Randomization Test: Exact Validity

Theorem (Exact Size Control)

Under the sharp null $H_0 : Y_{1,i} = Y_{0,i}$ for all i, for any $\alpha \in (0, 1)$, the randomization test that rejects when $p \leq \alpha$ satisfies

$$\Pr_{H_0}[\textit{reject } H_0] \leq \alpha.$$

- This holds *exactly* in finite samples, for *any* test statistic T, under *any* distribution of potential outcomes.
- The inequality is due to the discreteness of the randomization distribution; equality holds if α is an attainable significance level.

Choice of Test Statistic

Under the sharp null, the randomization test controls size for *any* test statistic. The choice of test statistic affects *power*. Common choices (for two-sided alternatives):

Test Statistic	Notes
$\left\lvert \overline{Y}_1 - \overline{Y}_0 \right\rvert$	Simple difference in means
$\left\lvert \frac{\overline{Y}_1 - \overline{Y}_0}{\hat{\sigma}_{\overline{Y}_1 - \overline{Y}_0}} \right\rvert$	Studentized; also controls size for H_0^W (see below).
Wilcoxon rank-sum	Better power with heavy-tailed outcomes
KS statistic	Tests for distributional differences

Properties of Randomization Tests

- **Exact finite-sample validity:** Size control is exact for any N, not asymptotic.
- **No distributional assumptions:** No assumption on the distribution of potential outcomes (nonparametric). We did not impose or need independence across units, or any distribution assumption on outcomes.
  - We do require well-defined potential outcomes under the design (e.g., no interference, or partial interference at the cluster level if villages are clusters).
- **No asymptotic approximation.**

Properties of Randomization Tests (cont'd)

- **Hypothesis being tested:** sharp null that $Y_{1,i} = Y_{0,i}$ for all i
  - The sharp null is a *stronger* hypothesis than $\frac{1}{N}\sum_i \left(Y_{1,i} - Y_{0,i}\right) = 0$.
  - Rejection of sharp null $\implies$ some unit i has $Y_{1,i} \neq Y_{0,i}$, but does not directly say anything about $\frac{1}{N}\sum_i \left(Y_{1,i} - Y_{0,i}\right)$.

- **Conditioning:** Inference is conditional on the set of units in the experiment and their potential outcomes—we are testing whether the treatment had any effect on the specific units in our study.

- **Generalization:** Extends to other sharp null hypotheses of form $H_0 : Y_{1,i} - Y_{0,i} = \tau_0$ for all i (constant treatment effect τ_0).

Sharp Null vs. Weak Null: Interpretation

- The randomization test has exact size control under the *sharp null* $H_0^S : Y_{1,i} = Y_{0,i}$ for all i.

- In practice, researchers often *interpret* rejection as evidence against the *weak null* (null of no effect on average):

$$H_0^W : \frac{1}{N} \sum_i \left(Y_{1,i} - Y_{0,i} \right) = 0.$$

- Since $H_0^S \implies H_0^W$, rejection of H_0^S is *not* the same as rejection of H_0^W. The sharp null can be false (heterogeneous effects that cancel) while the weak null is true.

Sharp Null vs. Weak Null: Using Randomization Tests for H_0^W

- The randomization reference distribution is computed *as if* the sharp null holds (since we need $Y_{1,i} = Y_{0,i}$ for all i to construct it). This reference distribution need not hold under H_0^W with heterogeneous effects that cancel.
- **Unstudentized statistic** $\left|\hat{\tau}\right|$: the randomization test may *not* control size for H_0^W, even asymptotically. Under the sharp null the reference distribution reflects a common outcome variance, while under H_0^W with heterogeneous effects the actual variance depends on the group-specific variances S_1^2, S_0^2 and the unobservable S_τ^2—these can diverge, causing over- or under-rejection.
- **Studentized statistic** $\left|\hat{\tau}\right| / \hat{\text{se}}$: studentizing ensures the randomization distribution and the actual sampling distribution under H_0^W converge to the same limit, so the test *also* controls size asymptotically for H_0^W under regularity conditions (finite second moments / Lindeberg conditions).
- See Wu and Ding (2021) for the finite-population/design-based analysis, and Ritzwoller, Romano, and Shaikh (2024) for the analogous result in the superpopulation (two-sample permutation test) framework.

Sharp Null vs. Weak Null (cont'd)

- **Summary of size control properties:**

	H_0^S **(sharp)**	H_0^W **(weak)**
Rand. test, any statistic	Exact, finite-sample	Not guaranteed
Rand. test, studentized	Exact, finite-sample	Asymptotic
Standard t-test	Asymptotic only	Asymptotic

- The studentized randomization test gives the **best of both worlds**: exact finite-sample size control for H_0^S, and asymptotic size control for H_0^W (under regularity conditions).
- **Practical recommendation:**
  - If the goal is to test the sharp null, the randomization test is exact for *any* test statistic.
  - If the goal is also to have validity for the weak null, use the *studentized* test statistic.
- See Wu and Ding (2021) and Ritzwoller, Romano, and Shaikh (2024) for further discussion.

Randomization Inference, Imposing Sharp Null

		Possible Treatment Assignments									
i	Y_i	$\mathbf{d}^{(1)}$	$\mathbf{d}^{(2)}$	$\mathbf{d}^{(3)}$	$\mathbf{d}^{(4)}$	$\mathbf{d}^{(5)}$	$\mathbf{d}^{(6)}$	$\mathbf{d}^{(7)}$	$\mathbf{d}^{(8)}$	$\mathbf{d}^{(9)}$	$\mathbf{d}^{(10)}$
1	0.65	1	**1**	1	1	1	1	0	0	0	0
2	0.62	1	**1**	1	0	0	0	1	1	1	0
3	0.81	1	**0**	0	1	1	0	1	1	0	1
4	0.88	0	**1**	0	1	0	1	1	0	1	1
5	0.60	0	**0**	1	0	1	1	0	1	1	1
$T = \left\lvert \overline{Y}_1 - \overline{Y}_0 \right\rvert =$		0.047	**0.012**	0.222	0.170	0.063	0.005	0.145	0.088	0.030	0.128

Actual treatment assignment and test statistic in bold.

- Showing the $\binom{5}{3} = 10$ possible ways to assign treatment to 3 out of 5 villages.
- Showing value of test statistics under null for each way to assign treatment.
- Re-ordering columns by magnitude of test statistic. . .

Randomization Inference, Imposing Sharp Null

		Possible Treatment Assignments									
i	Y_i	$\mathbf{d}^{(6)}$	$\mathbf{d}^{(2)}$	$\mathbf{d}^{(9)}$	$\mathbf{d}^{(1)}$	$\mathbf{d}^{(5)}$	$\mathbf{d}^{(8)}$	$\mathbf{d}^{(10)}$	$\mathbf{d}^{(7)}$	$\mathbf{d}^{(4)}$	$\mathbf{d}^{(3)}$
1	0.65	1	**1**	0	1	1	0	0	0	1	1
2	0.62	0	**1**	1	1	0	1	0	1	0	1
3	0.81	0	**0**	0	1	1	1	1	1	1	0
4	0.88	1	**1**	1	0	0	0	1	1	1	0
5	0.60	1	**0**	1	0	1	1	1	0	0	1
$T = \lvert\bar{Y}_1 - \bar{Y}_0\rvert =$		0.005	**0.012**	0.030	0.047	0.063	0.088	0.128	0.145	0.170	0.222

Actual treatment assignment and test statistic in bold. Reported values of T are under the sharp null of no effect.

- Under the null, for any t, $\Pr[T \geq t] = \frac{1}{10}\sum_{j=1}^{10} \mathbf{1}[T_j \geq t]$.
- For example,

$$\Pr[T \geq 0.05] = \frac{1}{10}(0 + 0 + 0 + 0 + 1 + \ldots + 1) = 0.6$$

Randomization Inference, Imposing Sharp Null

		Possible Treatment Assignments									
i	Y_i	$\mathbf{d}^{(6)}$	$\mathbf{d}^{(2)}$	$\mathbf{d}^{(9)}$	$\mathbf{d}^{(1)}$	$\mathbf{d}^{(5)}$	$\mathbf{d}^{(8)}$	$\mathbf{d}^{(10)}$	$\mathbf{d}^{(7)}$	$\mathbf{d}^{(4)}$	$\mathbf{d}^{(3)}$
1	0.65	1	**1**	0	1	1	0	0	0	1	1
2	0.62	0	**1**	1	1	0	1	0	1	0	1
3	0.81	0	**0**	0	1	1	1	1	1	1	0
4	0.88	1	**1**	1	0	0	0	1	1	1	0
5	0.60	1	**0**	1	0	1	1	1	0	0	1
$T = \lvert\bar{Y}_1 - \bar{Y}_0\rvert =$		0.005	**0.012**	0.030	0.047	0.063	0.088	0.128	0.145	0.170	0.222

Actual treatment assignment and test statistic in bold. Reported values of T are under the sharp null of no effect.

- P-value is fraction of test statistics at least as big as realized value 0.012:

$$\Pr[T \geq 0.012] = \frac{1}{10}(0 + 1 + \ldots + 1) = 0.9.$$

We fail to reject null . . .

Individual Child as Unit, Randomization at Village Level?

Take unit of observation to be individual child?

- We have been treating village as unit, but we actually want individual child as unit, though with randomization at village level (each village is a *cluster*).
- We can proceed as before, defining outcomes at individual child level, defining test statistic as averages over individual level outcomes, but still consider possible treatment assignments at village level.

Preview: Studentized Statistic with Clustered Assignment

- In PROGRESA, treatment is randomly assigned at the village level, not at the individual-child level.
- For testing the **sharp null**, randomization inference is exact for *any* test statistic, as long as we re-randomize treatment at the village level to match the actual design.
- For testing the **weak null**, studentization is important for asymptotically valid randomization inference. In the PROGRESA context, the standard error used in the studentized statistic should reflect that random assignment was at the village level.
- In this lecture, for simplicity, we use the unstudentized test statistic given by the absolute value of the child-weighted difference in means.
- In the next lecture on multiple hypothesis testing, we will consider the corresponding studentized statistic. Studentization has an additional advantage in multiple hypothesis testing that it makes the test statistics more comparable across hypotheses.

Randomization Inference Full Sample

- We have been considering 5 villages, but in full RCT we have 308 treated out of 491 villages.

```
> length(unique(dfPost$sooloca))
[1] 491
> length(unique(subset(dfPost,dfPost$progresa1==1)$sooloca))
[1] 308
```

Randomization Inference Full Sample

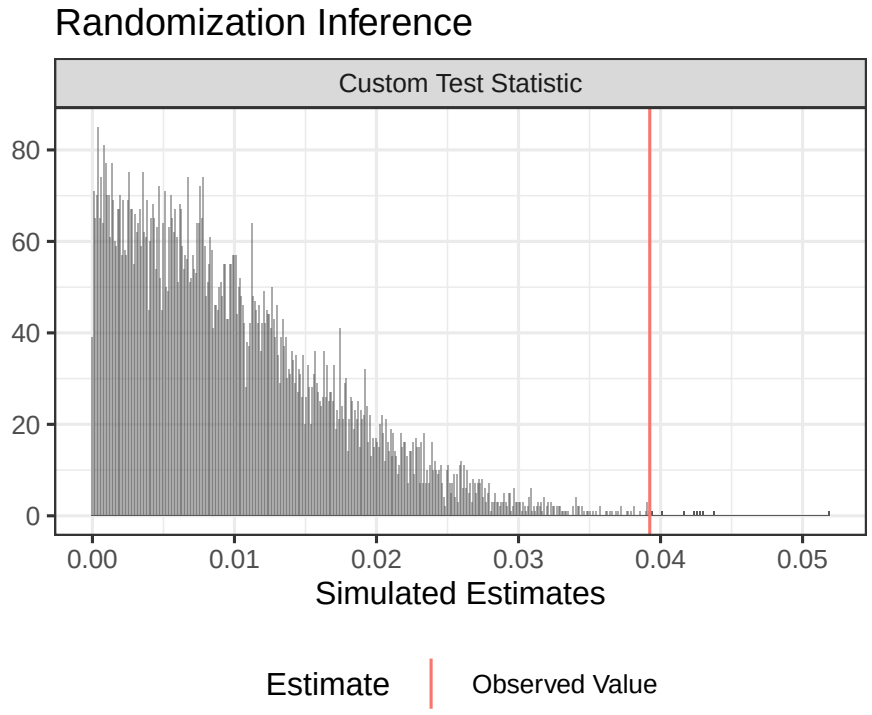
- We have been considering 5 villages, but in full RCT we have 308 treated out of 491 villages.
- In theory could proceed as before using full sample, enumerate all possible treatment assignments for 308 out of 491 units, but $\binom{491}{308} \approx 2.5 \times 10^{139}$—a number with 140 digits, far exceeding the estimated number of atoms in the observable universe ($\approx 10^{80}$). Complete enumeration is utterly infeasible.

```
> choose(491,308)
[1] 2.452172e+139
```

Randomization Inference Full Sample

- We have been considering 5 villages, but in full RCT we have 308 treated out of 491 villages.
- In theory could proceed as before using full sample, enumerate all possible treatment assignments for 308 out of 491 units, but $\binom{491}{308} \approx 2.5 \times 10^{139}$—a number with 140 digits, far exceeding the estimated number of atoms in the observable universe ($\approx 10^{80}$). Complete enumeration is utterly infeasible.
- Therefore, use stochastic approximation, randomly draw K out of the $\binom{491}{308}$ possible treatment assignments, proceed as before using the distribution of the test statistic over those K simulations. p-value is proportion of simulated test statistics at least as large as observed test statistic.
  - The resulting inference is using an approximation, justified as $K \to \infty$, though we choose K, only limited by our computing power.

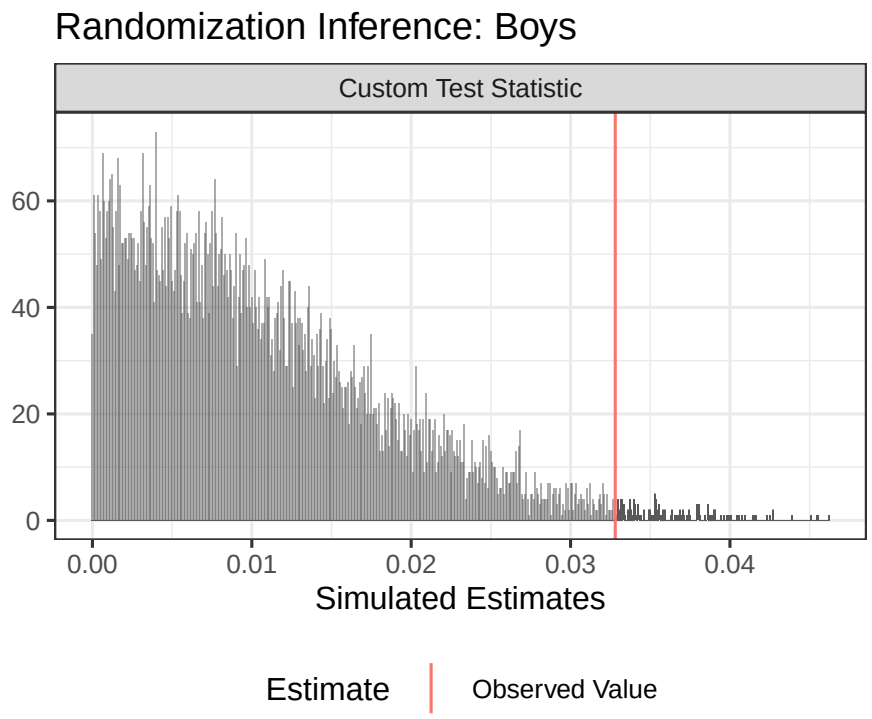
Randomization Inference on Sharp Null, Full Sample



Randomization inference on full PROGRESA Sample

- Using $3,000$ simulations from the $\binom{491}{308}$ possible treatment assignments.
- Implemented using package `ri2`.
- simulated p-value for sharp null of no effect: 0.0009.

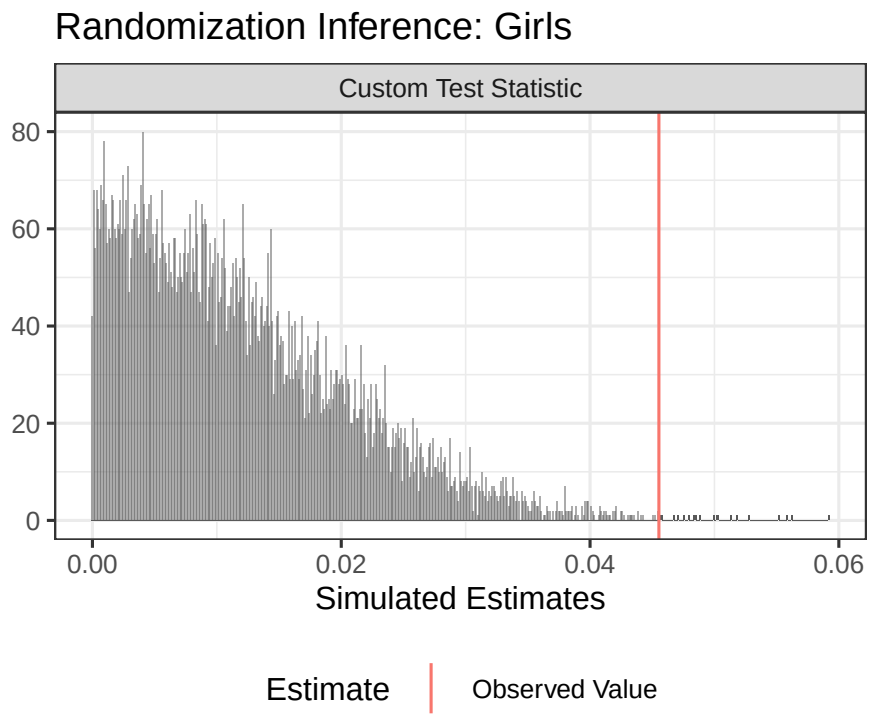
Randomization Inference on Sharp Null, by Sex of Child



Randomization inference conditional on sex of child:

- Using $3,000$ simulations from the $\binom{491}{308}$ possible treatment assignments.
- Implemented using package `ri2`.
- simulated p-value for sharp null of no effect:
  - For boys: 0.0125.

Randomization Inference on Sharp Null, by Sex of Child



Randomization inference conditional on sex of child:

- Using $3,000$ simulations from the $\binom{491}{308}$ possible treatment assignments.
- Implemented using package `ri2`.
- simulated p-value for sharp null of no effect:
  - For boys: 0.0125.
  - For girls: 0.002.

Randomization Inference on Sharp Null, by Sex and Grade of Child

	p-values	
Grade	boys	girls
1	0.347	0.609
2	0.005	0.931
3	0.714	0.339
4	0.0004	0.025
5	0.297	0.076
6	0.002	0.007
7	0.012	0.0001
8	0.558	0.628
9	0.276	0.681
10	0.267	0.657

Randomization inference conditional on sex and grade of child:

- Using $3,000$ simulations from the $\binom{491}{308}$ possible treatment assignments.
- Showing simulated p-value for sharp null of no effect.
- Correct for multiple hypothesis testing?
  (we will cover this next lecture)
- How would you summarize the evidence of whether PROGRESA has an effect, and on whom?

Digression: Re-randomization

- Practitioners often check baseline covariate balance between treated and control groups, and re-randomize if balance is not sufficiently achieved.
  - Bruhn and McKenzie (2009) documented that this was widespread in development field experiments, often without being reported.

Digression: Re-randomization (cont'd)

- Re-randomization will typically improve asymptotic precision (similar to proper regression adjustment for baseline differences), but ignoring the re-randomization procedure in the analysis invalidates conventional standard errors and inference.
- For example, Li, Ding and Rubin (2018, PNAS) show that if re-randomize based on Mahalanobis distance, then:
  - improves asymptotic precision to almost as good as linear regression adjustment,
    - Compare results to Lin (2013) (design-based), see also Negi and Wooldridge (2021) (superpopulation, i.i.d. assignment).
  - conventional standard errors, inference are not valid;
  - Exploiting known criteria for re-randomization, can adjust standard errors, inference to be asymptotically valid.

Re-randomization: Formal Setup

- Let $\phi(\mathbf{d})$ be an acceptance function: re-randomize until $\phi(\mathbf{D}) = 1$.
- Morgan and Rubin (2012) formalize: the effective randomization distribution is

$$\Pr_{\text{ReRand}}[\mathbf{D} = \mathbf{d}] = \frac{\mathbf{1}[\phi(\mathbf{d}) = 1]}{\sum_{\mathbf{d}' \in \Omega} \mathbf{1}[\phi(\mathbf{d}') = 1]}.$$

- Common criterion: accept if the Mahalanobis distance $M = (\bar{\mathbf{x}}_T - \bar{\mathbf{x}}_C)' \hat{\mathbf{S}}^{-1} (\bar{\mathbf{x}}_T - \bar{\mathbf{x}}_C) \leq a$ for some threshold a, where $\hat{\mathbf{S}}$ is the pooled sample covariance matrix of covariates. This is a scale- and correlation-adjusted measure of how far apart the treatment and control covariate means are.

Re-randomization: Formal Setup (cont'd)

- Li, Ding and Rubin (2018): under re-randomization with $M \leq a$,
  - the difference-in-means estimator is unbiased (as always under complete randomization);
  - its variance is reduced, with the reduction depending on the R^2 of outcomes on covariates;
  - asymptotic gains from re-randomization approach those from regression adjustment as $a \longrightarrow 0$.
  - The usual Neyman variance estimator is conservative, so standard inference remains valid (but less powerful).

Re-randomization and Randomization Inference

- Randomization inference remains valid with re-randomization if we exploit the known re-randomization criteria.
  - The randomization distribution is now over the *restricted* set of acceptable assignments:

$$\Pr_{H_0}[T \geq t] = \frac{\sum_{\mathbf{d} \in \Omega_a} \mathbf{1}[T(\mathbf{Y}, \mathbf{d}) \geq t]}{|\Omega_a|}, \qquad \Omega_a = \{\mathbf{d} \in \Omega : \phi(\mathbf{d}) = 1\}.$$

  - When using stochastic approximation, simulate treatment assignments from Ω_a (draw and accept/reject).

- General principle: the reference distribution for randomization inference should reflect the *actual* randomization procedure used.

Post-Randomization Covariate Adjustment

Alternative to re-randomization: adjust *after* randomization via regression.

- **Freedman (2008):** Considers the *additive* specification $Y_i = \alpha + \beta D_i + \gamma' W_i + \varepsilon_i$ (common covariate slopes across treatment and control). Shows this estimator can be biased in finite samples, that adjustment may *increase* asymptotic variance relative to the unadjusted difference-in-means, and that conventional OLS standard errors are invalid.
- **Lin (2013):** Responding to Freedman, shows that allowing *separate* covariate slopes via the *interacted* specification—OLS of Y_i on D_i, $\tilde{W}_i$, and $D_i \cdot \tilde{W}_i$ (where $\tilde{W}_i = W_i - \bar{W}$)—resolves these problems in a *completely randomized* experiment:
  - a consistent estimator of the ATE (but not necessarily unbiased in finite samples, unlike the unadjusted difference-in-means),
  - asymptotic variance no larger than that of the unadjusted difference-in-means—strictly smaller when covariates predict outcomes,
  - heteroskedasticity-robust standard errors are asymptotically valid, even if the linear model is misspecified.

Post-Randomization Covariate Adjustment (cont'd)

- Chang, Middleton, and Aronow (2024) derive exact finite-sample bias corrections for regression adjustment, resolving Freedman's concerns with minor modifications to practice.
- Freedman (2008), Lin (2013), and Chang, Middleton, and Aronow (2024) are all *design-based*. In contrast, Negi and Wooldridge (2021) justify the same interacted regression as Lin in a *superpopulation* framework with *i.i.d. treatment assignment* (simple random assignment, not CRE), with heteroskedasticity-robust standard errors asymptotically valid regardless of correct specification.

Post-Randomization Covariate Adjustment (cont'd)

- Ma, Tu, and Liu (2022) extend to a superpopulation framework with *covariate-adaptive randomization* (including CRE and stratified designs, see below), using the Bugni, Canay, and Shaikh (2018) conditions on the assignment mechanism. They show that the interacted regression is *consistent* for the ATE regardless of model misspecification, and characterize when the OLS variance estimator suffices vs. when a nonparametric variance estimator is needed. Note: consistency does not imply the "no harm" efficiency property—under stratification, the interacted regression can be strictly less efficient than the unadjusted difference-in-means (see Cytrynbaum, 2024, below).

Covariate-Adaptive Randomization

- **Covariate-adaptive randomization (CAR):** Treatment assignment probabilities depend on covariates (and possibly previous assignments) to promote balance. Includes:
  - **Stratified (blocked) randomization:** Partition units into strata based on baseline covariates; randomize within each stratum with fixed treatment counts.
  - **Biased coin designs:** Assignment probabilities tilt toward whichever treatment group is currently underrepresented given the incoming unit's covariates, but remain strictly between 0 and 1.

- Bugni, Canay, and Shaikh (2018): show conventional t-test is conservative under CAR; propose valid inference. Bugni, Canay, and Shaikh (2019) extends to multiple treatments. Analysis is in a *superpopulation* framework, but carefully accounts for the randomization procedure induced by CAR.

Covariate-Adaptive Randomization (cont'd)

- Bai (2022): shows a specific matched-pair design achieves maximum precision for ATE estimation among all stratified randomization schemes; related work (Bai, Romano, and Shaikh, 2022). Analysis is in a *superpopulation* framework, but carefully accounts for the randomization procedure induced by CAR.

- Cytrynbaum (2024) shows Lin's result does *not* generalize to stratified experiments—standard regression adjustment with treatment-covariate interactions can be *strictly inefficient*, potentially increasing variance relative to the unadjusted difference in means. Derives the optimal linear covariate adjustment for a given stratification.

- Cytrynbaum (2024): introduces a family of "local randomization" designs that unify stratified block randomization and matched pairs. Solves the optimal stratification problem with heterogeneous costs; provides asymptotically exact inference methods that fully exploit precision gains from stratified sampling and assignment.

Implementation in R

- **Randomization inference** (Fisher randomization tests):
  - `ri2` (Coppock), building on the original `ri` package (Aronow and Samii): Specify the randomization procedure via the `randomizr` package, choose a test statistic, and simulate the randomization distribution under a sharp null. Supports simple, complete, blocked, clustered, and block-clustered designs.
- **Estimation and asymptotic inference** in stratified/clustered experiments:
  - `sreg` (Trifonov, Bai, Shaikh, and Tabord-Meehan): Computes ATE estimates with valid standard errors and confidence intervals for stratified block randomization (including matched pairs and k-tuples), multiple treatments, cluster randomization, and optimal covariate adjustment. Companion package to Bai, Shaikh, and Tabord-Meehan (2025).
- Note: `ri2` and `sreg` are **complementary**, not substitutes. `ri2` constructs exact p-values by simulation under sharp nulls; `sreg` provides asymptotic inference with correct variance estimation under stratification.

Current Research

- Design and analysis of randomized experiments is an **active area of current research**.
- Selected recent topics include:
  - Optimal stratification and finely stratified designs (Bai, 2022; Bai, Liu, Shaikh, and Tabord-Meehan, 2023; Cytrynbaum, 2024a; Cytrynbaum, 2024b)
  - Inference under covariate-adaptive randomization (Bugni, Canay, and Shaikh, 2018, 2019)
  - Rerandomization and regression adjustment (Li and Ding, 2020; Zhao and Ding, 2021)
  - Exact inference and bias correction (Chang, Middleton, and Aronow, 2024)
  - Experiments under interference (Aronow and Samii, 2017; Athey, Eckles, and Imbens, 2018)
  - Cluster randomization (Bugni, Canay, Shaikh, and Tabord-Meehan, 2025)
- **Overview paper:** Bai, Shaikh, and Tabord-Meehan (2025).